

Making Sense of Complexity with Katy Börner: Constructing and Exploring Data Visualizations of Complex Systems



📅 **Date**

Wednesday, February 4, 2026

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12:15 p.m. to 1:15 p.m.

📍 Location

Engineering Center A (ECA) 100
[1031 Palm Walk Tempe AZ 85281](#)

💰 Cost

Free

Register here

Event description

Free

Open to the public

Science

Sustainability

Envisioning and implementing desirable futures requires a deep understanding of developments in science and technology as well as the ability to both simulate and communicate the likely impact of alternative actions. At a time when our relationship to a vulnerable planet Earth is especially important, such a profound awareness of complex, interlinked systems is needed more than ever. Atlas of Forecasts, from the creator of Atlas of Science and Atlas of Knowledge shows how we can use data to map possible futures.

Börner's Atlas of Forecasts uses advanced data visualizations to introduce different types of computational models, and it demonstrates how model results can be used to inform effective decision-making. The models aim to capture the structure and dynamics of developments in education and the job market, progress in science and technology, and the impact of government policies—all from the micro to the macro levels. Model results can help us decide which human skills are needed in an artificial intelligence—empowered economy; which courses and degrees are most effective in upskilling and reskilling the current and future workforce; what progress in science and technology is likely to happen; and how policymakers can future-proof regions or nations. This Atlas offers a driver's seat-perspective for a test-drive of the future.

End of 2025, the Atlas of Macroscopes: Interactive Data Visualizations became available in stores around the globe. Transcending the static nature of the maps featured in the initial trilogy, macroscopes, or interactive data visualizations, give us holistic views of complex systems or networks. They provide an entry point for scientists and laypeople alike and empower us to engage directly with large datasets and to conduct our own lines of questioning. As portals to continuously evolving data, macroscopes

Biography

KATY BÖRNER is the Victor H. Yngve Distinguished Professor of Engineering and Information Science in the Departments of Intelligent Systems Engineering and Information Science, Luddy School of Informatics, Computing, and Engineering; core faculty of the Cognitive Science Program; and founding director of the Cyberinfrastructure for Network Science Center (<http://cns.iu.edu>) —all at Indiana University in Bloomington, Indiana. She is a curator of the Mapping Science exhibit (<http://scimaps.org>) and she leads the international Human Reference Atlas effort (<https://humanatlas.io>). She holds an MS in electrical engineering from the University of Technology in Leipzig, and a PhD in computer science from the University of Kaiserslautern. Börner is a Fellow of the American Association for the Advancement of Science (AAAS), a Humboldt Research Fellow, an Association for Computing Machinery (ACM) Fellow, Network Society Fellow, and a Stiftung Charité Visiting Fellow.

References

- Börner, Katy, Philip Blood, Jonathan C. Silverstein, et al. 2025. "[Human BioMolecular Atlas Program \(HuBMAP\): 3D Human Reference Atlas Construction and Usage](#)". Nature Methods. doi: 10.1038/s41592-024-02563-5
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- Börner, Katy. 2015. [Atlas of Knowledge: Anyone Can Map](#). Cambridge, MA: The MIT Press.
- Börner, Katy. 2010. [Atlas of Science: Visualizing What We Know](#). Cambridge, MA: MIT Press.

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